

NAGASAKI, Japan

WMO: 478170

Lat: 32.7343N

Lon: 129.8680E

Elev: 118

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.4	35.3	18.9	14.3	40.9	21.3	16.0	40.9	19.8	47.4	17.5	49.3	4.9	80	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	91.4	78.4	89.6	78.3	87.9	77.7	80.7	86.9	79.8	85.9	79.1	85.0	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.5	83.8	78.2	147.3	83.4	77.5	143.4	83.0	44.5	87.2	43.5	86.0	42.8	84.9	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	14.4	12.3	29.8	95.8	1.8	1.8	28.5	97.1	27.5	98.1	26.5	99.2	25.2	100.5
			26.9	81.4	1.7	1.2	25.6	82.2	24.6	82.9	23.6	83.6	22.4	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	44.8	46.7	52.2	60.3	67.9	73.8	81.2	83.4	77.6	68.8	58.4	49.0
	DBStd	13.91	4.99	5.67	5.46	4.93	3.50	3.40	3.54	3.02	4.08	4.71	5.72	5.47
	HDD50	425	174	122	38	1	0	0	0	0	0	0	4	86
	HDD65	2420	626	511	396	153	13	0	0	0	0	18	207	496
	CDD50	5455	13	31	107	310	554	714	968	1035	829	582	257	55
	CDD65	1973	0	0	0	12	101	264	503	570	379	134	10	0
	CDH74	16676	0	0	0	19	246	1212	5152	6619	2950	472	6	0
	CDH80	5203	0	0	0	1	12	140	1658	2606	740	46	0	0
Wind	WSAvg	5.1	5.1	5.4	5.8	5.5	4.9	5.0	5.7	5.0	4.8	4.6	4.5	4.9
Precipitation	PrecAvg	75.6	2.8	3.3	4.7	6.6	7.1	13.0	11.8	7.8	7.4	4.1	3.7	2.8
	PrecMax	112.0	9.3	7.6	13.8	13.7	15.4	31.1	46.4	26.3	19.3	11.0	8.4	6.0
	PrecMin	36.4	0.4	0.9	1.7	1.4	2.0	1.7	0.2	0.3	0.1	0.2	0.5	0.2
	PrecStd	15.4	1.5	1.7	2.0	2.6	3.1	6.2	8.5	5.4	4.3	2.8	2.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	66.4	70.1	76.4	81.7	85.8	92.9	95.0	90.5	84.5	75.0	65.9
		MCWB	57.6	59.7	59.9	64.2	68.0	74.5	78.2	78.5	76.7	72.4	66.1	59.8
	2%	DB	58.8	62.9	66.8	73.1	78.7	83.2	90.2	92.3	88.1	80.9	72.0	63.0
		MCWB	53.4	57.6	58.3	62.3	66.4	73.7	78.7	78.3	76.4	70.1	63.7	55.9
	5%	DB	55.8	59.6	64.2	71.0	76.6	81.5	88.6	90.6	86.1	78.7	69.6	60.4
		MCWB	49.3	53.7	56.8	61.5	65.9	73.4	78.3	78.4	75.1	68.2	62.4	53.6
	10%	DB	53.1	56.3	61.7	68.7	74.9	79.9	87.0	89.0	84.2	76.5	67.5	57.9
		MCWB	46.7	49.6	54.9	60.2	65.2	72.7	77.8	78.2	74.3	66.8	60.5	51.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	62.1	63.7	69.7	73.5	78.6	82.0	81.9	80.0	76.5	69.1	62.8
		MCDB	61.9	64.2	66.6	72.9	77.0	82.5	87.8	88.1	85.3	80.3	71.9	64.5
	2%	WB	54.7	58.7	60.7	66.4	71.0	77.4	80.6	80.9	78.7	73.7	66.5	57.7
		MCDB	57.8	62.0	64.6	70.0	74.3	81.1	86.3	87.4	84.0	78.5	69.6	61.1
	5%	WB	50.3	54.7	58.2	64.2	69.1	76.2	79.7	80.1	77.5	71.4	64.0	55.0
		MCDB	54.4	58.7	63.0	68.3	73.1	79.8	85.2	86.6	83.3	76.4	68.3	59.0
	10%	WB	47.6	50.8	55.8	62.0	67.3	74.7	78.9	79.5	76.4	68.7	61.7	52.5
		MCDB	52.2	55.0	61.0	67.0	72.5	78.1	84.3	85.9	82.3	74.2	66.4	57.1
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.2	12.8	13.2	12.7	10.0	9.2	10.3	11.1	12.4	12.2	11.6
		MCDBR	14.6	15.7	15.8	15.9	15.1	12.1	11.3	12.2	12.5	14.1	13.4	14.3
		MCWBR	11.1	12.0	10.9	8.9	7.7	5.7	4.4	4.4	5.2	6.7	8.3	9.9
	5% WB	MCDBR	13.6	13.9	13.4	11.9	10.9	8.0	9.6	10.4	10.5	10.8	11.1	12.6
		MCWBR	12.1	13.1	11.8	9.4	7.9	5.0	4.4	4.4	5.1	6.9	8.9	10.8
Clear-Sky Solar Irradiance	taub	0.445	0.484	0.543	0.562	0.571	0.582	0.564	0.539	0.512	0.468	0.438	0.434	
	taud	2.025	1.959	1.822	1.830	1.870	1.932	2.019	2.091	2.113	2.135	2.133	2.064	
	Ebn at Noon	234	239	235	238	237	233	237	241	241	242	236	229	
	Edh at Noon	44	52	64	67	64	60	55	51	48	43	39	41	
All-Sky Solar Radiation	RadAvg	687	929	1277	1536	1634	1373	1639	1706	1384	1151	827	641	
	RadStd	70	92	84	132	187	213	225	150	141	113	82	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page